

ASSIGNMENT NO.4: Working with Multiple Tables and Joins

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Questions:

Q1) Create a MySQL database named FY_CollegeDB and create the following tables:

Student Table

Column Name	Data Type	Constraint
StudentID	INT	PRIMARY KEY
StudentName	VARCHAR(50)	
Column Name	Data Type	Constraint
City	VARCHAR(30)	

Course Table

Column Name	Data Type	Constraint
CourseID	INT	PRIMARY KEY
CourseName	VARCHAR(50)	
Fees	DECIMAL(10,2)	

Enrollment Table

Column Name	Data Type	Constraint
EnrollmentID	INT	PRIMARY KEY
StudentID	INT	FOREIGN KEY
CourseID	INT	FOREIGN KEY
EnrollmentDate	DATE	

QUERY:

```

1 CREATE DATABASE FY_CollegeDB;
2 USE FY_CollegeDB;
3
4 CREATE TABLE Student (
5     StudentID INT PRIMARY KEY,
6     StudentName VARCHAR(50),
7     City VARCHAR(30)
8 );
9
10 CREATE TABLE Course (
11     CourseID INT PRIMARY KEY,
12     CourseName VARCHAR(50),
13     Fees DECIMAL(10,2)
14 );
15
16 CREATE TABLE Enrollment (
17     EnrollmentID INT PRIMARY KEY,
18     StudentID INT,
19     CourseID INT,
20     EnrollmentDate DATE,
21     FOREIGN KEY (StudentID) REFERENCES Student(StudentID),
22     FOREIGN KEY (CourseID) REFERENCES Course(CourseID)
23 );

```

OUTPUT:

Tables_in_fy_collegedb
course
enrollment
student

Q2) Insert at least 6 records into the Student table, 4 records into the Course table, and 5 records into the Enrollment table.

Make sure:

Requirement
At least one student should not have enrollment
At least one course should not have any enrolled student

QUERY:

```

1  USE FY_CollegeDB;
2
3  DELETE FROM Enrollment;
4  DELETE FROM Student;
5  DELETE FROM Course;
6
7  INSERT INTO Student (StudentID, StudentName, City)
8  VALUES
9  (1, 'Aarav Shah', 'Mumbai'),
10 (2, 'Meera Joshi', 'Pune'),
11 (3, 'Rohan Patil', 'Nashik'),
12 (4, 'Sneha Desai', 'Mumbai'),
13 (5, 'Amit Kumar', 'Delhi'),
14 (6, 'Neha Sharma', 'Pune');
15
16 INSERT INTO Course (CourseID, CourseName, Fees)
17 VALUES
18 (101, 'Data Science', 75000.00),
19 (102, 'Computer Science', 70000.00),
20 (103, 'Cyber Security', 60000.00),
21 (104, 'Web Development', 45000.00);
22
23 INSERT INTO Enrollment (EnrollmentID, StudentID, CourseID, EnrollmentDate)
24 VALUES
25 (1, 1, 101, '2026-07-01'),
26 (2, 2, 102, '2026-07-02'),
27 (3, 3, 101, '2026-07-03'),
28 (4, 4, 103, '2026-07-04'),
29 (5, 5, 102, '2026-07-05');
30
31 SELECT * FROM Student;
32 SELECT * FROM Course;
33 SELECT * FROM Enrollment;

```

OUTPUT:

1 Student output :

	StudentID	StudentName	City
☐ Edit Copy Delete	1	Aarav Shah	Mumbai
☐ Edit Copy Delete	2	Meera Joshi	Pune
☐ Edit Copy Delete	3	Rohan Patil	Nashik
☐ Edit Copy Delete	4	Sneha Desai	Mumbai
☐ Edit Copy Delete	5	Amit Kumar	Delhi
☐ Edit Copy Delete	6	Neha Sharma	Pune

2 Course output:

				CourseID	CourseName	Fees			
		Edit		Copy		Delete	101	Data Science	75000.00
		Edit		Copy		Delete	102	Computer Science	70000.00
		Edit		Copy		Delete	103	Cyber Security	60000.00
		Edit		Copy		Delete	104	Web Development	45000.00

3 Enrollment output:

				EnrollmentID	StudentID	CourseID	EnrollmentDate
☐		Edit		Copy		Delete	
				1	1	101	2026-07-01
☐		Edit		Copy		Delete	
				2	2	102	2026-07-02
☐		Edit		Copy		Delete	
				3	3	101	2026-07-03
☐		Edit		Copy		Delete	
				4	4	103	2026-07-04
☐		Edit		Copy		Delete	
				5	5	102	2026-07-05

Q3) Write a MySQL query using INNER JOIN to display students who have enrolled in courses.

The output should include:

Column
StudentID
StudentName
CourseID
EnrollmentDate

QUERY:

```

1 SELECT Student.StudentID, Student.StudentName,
2       Enrollment.CourseID, Enrollment.EnrollmentDate
3 FROM Student
4 INNER JOIN Enrollment
5 ON Student.StudentID = Enrollment.StudentID;
```

OUTPUT:

StudentID	StudentName	CourseID	EnrollmentDate
1	Aarav Shah	101	2026-07-01
2	Meera Joshi	102	2026-07-02
3	Rohan Patil	101	2026-07-03
4	Sneha Desai	103	2026-07-04
5	Amit Kumar	102	2026-07-05

Q4) Write a MySQL query to display student name, city, course name, fees, and enrollment date using multiple tables. Use:

Tables Required
Student
Course
Enrollment

QUERY:

```

1 SELECT Student.StudentName, Student.City,
2       Course.CourseName, Course.Fees,
3       Enrollment.EnrollmentDate
4 FROM Student
5 INNER JOIN Enrollment
6 ON Student.StudentID = Enrollment.StudentID
7 INNER JOIN Course
8 ON Enrollment.CourseID = Course.CourseID;
```

OUTPUT:

StudentName	City	CourseName	Fees	EnrollmentDate
Aarav Shah	Mumbai	Data Science	75000.00	2026-07-01
Rohan Patil	Nashik	Data Science	75000.00	2026-07-03
Meera Joshi	Pune	Computer Science	70000.00	2026-07-02
Amit Kumar	Delhi	Computer Science	70000.00	2026-07-05
Sneha Desai	Mumbai	Cyber Security	60000.00	2026-07-04

Q5) Write a MySQL query using LEFT JOIN to display all students, including those who have not enrolled in any course. The output should include:

Column
StudentID
StudentName

City
CourseID
EnrollmentDate

QUERY:

```

1 SELECT Student.StudentID,
2       Student.StudentName,
3       Student.City,
4       Enrollment.CourseID,
5       Enrollment.EnrollmentDate
6 FROM Student
7 LEFT JOIN Enrollment
8 ON Student.StudentID = Enrollment.StudentID;

```

OUTPUT:

StudentID	StudentName	City	CourseID	EnrollmentDate
1	Aarav Shah	Mumbai	101	2026-07-01
2	Meera Joshi	Pune	102	2026-07-02
3	Rohan Patil	Nashik	101	2026-07-03
4	Sneha Desai	Mumbai	103	2026-07-04
5	Amit Kumar	Delhi	102	2026-07-05
6	Neha Sharma	Pune	NULL	NULL

Q6) Write a MySQL query using RIGHT JOIN to display all courses, including courses that have no enrolled students. The output should include:

Column
StudentName
CourseID
CourseName
Fees

QUERY:

```

1
2 SELECT Student.StudentName,
3         Course.CourseID,
4         Course.CourseName,
5         Course.Fees
6 FROM Enrollment
7 RIGHT JOIN Course
8 ON Enrollment.CourseID = Course.CourseID
9 LEFT JOIN Student
10 ON Enrollment.StudentID = Student.StudentID;

```

OUTPUT:

StudentName	CourseID	CourseName	Fees
Aarav Shah	101	Data Science	75000.00
Meera Joshi	102	Computer Science	70000.00
Rohan Patil	101	Data Science	75000.00
Sneha Desai	103	Cyber Security	60000.00
Amit Kumar	102	Computer Science	70000.00
NULL	104	Web Development	45000.00